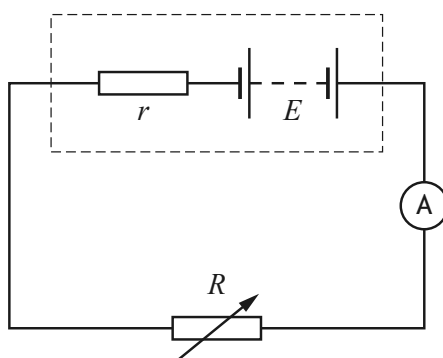
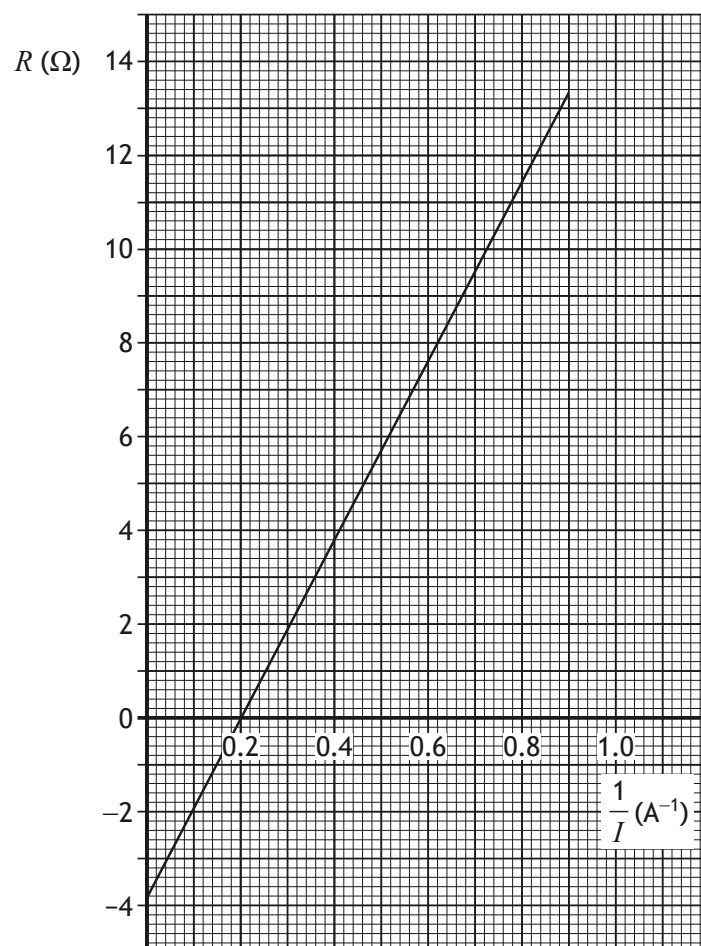


14. A technician sets up the following circuit.



The technician uses readings of resistance R and current I from the circuit to produce the graph shown.



* X 8 5 7 7 6 0 1 4 2 *

14. (continued)

Conservation of energy applied to the complete circuit gives the following relationship.

$$R = \frac{E}{I} - r$$

This relationship is in the form of the equation of a straight line.

$$y = mx + c$$

(a) Use the graph to determine:

(i) the internal resistance of the battery

1

(ii) the EMF of the battery

2

Space for working and answer

(iii) the short circuit current.

2

Space for working and answer

[Turn over



* X 8 5 7 7 6 0 1 4 3 *

14. (continued)

- (b) The technician now connects a second variable resistor in parallel with the original variable resistor.

State whether the short circuit current for this circuit will be greater than, less than, or the same as the value determined in (a) (iii).

Justify your answer.

2



* X 8 5 7 7 6 0 1 4 4 *